Configurations Derived from R, MTR, and D Configurations

This PDF provides the full specification of all configuration variants used in our experiments, derived from R, D, and MTR configurations. Tables 1–7 list variants obtained by changing exactly one setting from its baseline value in the corresponding base configuration while keeping all other settings unchanged.

Table 1: Characteristics of the *R+TurboBoost*, *D-TurboBoost*, and *MTR+TurboBoost* system configurations. Differences from the R, MTR, and D configurations are highlighted in blue.

Configuration	Turbo Boost	Scaling governors	C-states	Hyper-threading	Pin to NUMA node	Pin to isolated core	Fix start and max heap size	Fix young generation heap size	Process priority	Clear caches	Swap memory	NTP
D-TurboBoost	Off	Powersave	0-9	On	No	No	No	No	Default	No	On	On
R+TurboBoost	On	Performance	0	Off	Yes	Yes	Yes (16GB)	Yes (8GB)	Increased	Yes	Off	Off
MTR+TurboBoost	On	Powersave	0	On	Yes	No	Yes (16GB)	Yes (8GB)	Increased	Yes	Off	Off

Table 2: Characteristics of the *R+Powersave*, *R+Schedutil*, *D+Performance*, *MTR+Powersave*, and *MTR+Schedutil* system configurations. Differences from the R, MTR, and D configurations are highlighted in blue.

Configuration	Turbo Boost	Scaling governors	C-states	Hyper-threading	Pin to NUMA node	Pin to isolated core	Fix start and max heap size	Fix young generation heap size	Process priority	Clear caches	Swap memory	NTP
D+Performance	Off	Performance	0-9	On	No	No	No	No	Default	No	On	On
R+Powersave/R+Schedutil	On	Powersave/Schedutil	0	Off	Yes	Yes	Yes (16GB)	Yes (8GB)	Increased	Yes	Off	Off
MTR+Powersave/MTR+Schedutil	On	Powersave/Schedutil	0	On	Yes	No	Yes (16GB)	Yes (8GB)	Increased	Yes	Off	Off

Table 3: Characteristics of the $R{+}C0-9$, $D{+}C0$, and $MTR{+}C0-9$ system configurations. Differences from the R, MTR, and D configurations are highlighted in blue.

Configuration	Turbo Boost	Scaling governors	C-states	Hyper-threading	Pin to NUMA node	Pin to isolated core	Fix start and max heap size	Fix young generation heap size	Process priority	Clear caches	Swap memory	NTP
Default	On	Powersave	0	On	No	No	No	No	Default	No	On	On
Recommended	Off	Performance	$0-9$	Off	Yes	Yes	Yes (16GB)	Yes (8GB)	Increased	Yes	Off	Off
MT Recommended	Off	Powersave	$0-9$	On	Yes	No	Yes (16GB)	Yes (8GB)	Increased	Yes	Off	Off

Table 4: Characteristics of the *R+HTreading* and *D-HTreading* system configurations. Differences from the R and D configurations are highlighted in blue.

Configuration	Turbo Boost	Scaling governors	C-states	Hyper-threading	Pin to NUMA node	Pin to isolated core	Fix start and max heap size	Fix young generation heap size	Process priority	Clear caches	Swap memory	NTP
Default	On	Powersave	0-9	Off	No	No	No	No	Default	No	On	On
Recommended	Off	Performance	0	On	Yes	Yes	Yes (16GB)	Yes (8GB)	Increased	Yes	Off	Off

Table 5: Characteristics of the *R-Pinned* and *D+Pinned* system configurations. Differences from the R and D configurations are highlighted in red.

Configuration	Turbo Boost	Scaling governors	C-states	Hyper-threading	Pin to NUMA node	Pin to isolated core	Fix start and max heap size	Fix young generation heap size	Process priority	Clear caches	Swap memory	NTP
Default	On	Powersave	0-9	On	Yes	Yes	No	No	Default	No	On	On
Recommended	Off	Performance	0	Off	No	No	Yes (16GB)	Yes (8GB)	Increased	Yes	Off	Off
MT Recommended	Off	Powersave	0	On	Yes	No	Yes (16GB)	Yes (8GB)	Increased	Yes	Off	Off

Table 6: Characteristics of the *R+PriorityDef*, *D+PriorityInc*, and *MTR+PriorityDef* system configurations. Differences from the R, MTR, and D configurations are highlighted in blue.

Configuration	Turbo Boost	Scaling governors	C-states	Hyper-threading	Pin to NUMA node	Pin to isolated core	Fix start and max heap size	Fix young generation heap size	Process priority	Clear caches	Swap memory	NTP
Default	On	Powersave	$0-9$	On	No	No	No	No	Increased	No	On	On
Recommended	Off	Performance	0	Off	Yes	Yes	Yes (16GB)	Yes (8GB)	Default	Yes	Off	Off
MT Recommended	Off	Powersave	0	On	Yes	No	Yes (16GB)	Yes (8GB)	Default	Yes	Off	Off

Table 7: Characteristics of the *R-FixedHeap*, *D+FixedHeap*, and *MTR-FixedHeap* system configurations. Differences from the R, MTR, and D configurations are highlighted in blue.

Configuration	Turbo Boost	Scaling governors	C-states	Hyper-threading	Pin to NUMA node	Pin to isolated core	Fix start and max heap size	Fix young generation heap size	Process priority	Clear caches	Swap memory	NTP
Default	On	Powersave	$0-9$	On	No	No	Yes (16GB)	Yes (8GB)	Default	No	On	On
Recommended	Off	Performance	0	Off	Yes	Yes	No	No	Increased	Yes	Off	Off
MT Recommended	Off	Powersave	0	On	Yes	No	No	No	Increased	Yes	Off	Off